Sales Performance Analysis

1. Project Overview

This project analyzes retail sales transaction data to identify trends that improve revenue, profitability, and overall business performance. It focuses on understanding how factors such as product demand, pricing differences, regional performance, and seasonal trends influence sales outcomes.

The analysis aims to uncover key insights related to revenue growth, product performance, and profit optimization, supporting data-driven decision-making and strategic planning.

2. Dataset Summary

-Rows :200

-Columns:14

Key Features:

- Sales details (Order trends, Revenue performance, Units sold, Time-based patterns)
- Product insights (Product category performance, Pricing analysis, Profit margins)
- Regional distribution (State-wise sales, Region-wise revenue contribution)
- Profitability metrics (Cost vs. revenue comparison, Margin analysis, High-performing segments)

-Missing Data: 0 values

3. Exploratory Data Analysis (EDA) – Python

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

	OrderID	Year	UnitsSold	UnitCost	UnitPrice	ProductCost	SalesAmount
count	200.000000	200.0	200.000000	200.000000	200.000000	200.000000	200.000000
mean	3100.500000	2024.0	3.460000	12375.500000	17437.500000	40556.000000	57592.500000
std	57.879185	0.0	1.747475	11605.086814	14529.119738	41023.406286	53528.583392
min	3001.000000	2024.0	1.000000	500.000000	1000.000000	500.000000	1000.000000
25%	3050.750000	2024.0	2.000000	2000.000000	3500.000000	9000.000000	15000.000000
50%	3100.500000	2024.0	3.000000	12000.000000	18000.000000	32000.000000	44000.000000
75%	3150.250000	2024.0	5.000000	16000.000000	22000.000000	61000.000000	90000.000000
max	3200.000000	2024.0	6.000000	48000.000000	60000.000000	288000.000000	360000.000000

Missing Value Treatment

• Identified missing values using df.isnull().sum().

Feature Engineering

• Created a new feature **Profit** to calculate the difference between revenue and total cost, helping measure actual earnings from each transaction.

• Created **Profit (%)** to determine the profit margin, enabling better comparison of profitability across products, regions, and time periods.

Column Standardization

• Removed leading and trailing spaces from column names to ensure cleanliness and avoid referencing errors.

• Maintained consistent and clear naming conventions to support smooth analysis in Python, SQL, and visualization tools.

Database Integration

• Exported cleaned data to SQL database.
• Performed queries to analyze total revenue and overall profit performance.
• Identified top-performing products based on sales and determined which category contributes the highest profit..

4. Sales and Profit Analysis Using SQL

Using MySQL Workbench, we executed structured SQL queries to analyze total revenue, product-wise sales performance, and category-wise profit contribution.

1.Total Revenue and Profit Analysis – Calculated the overall revenue generated and total profit earned to evaluate overall business performance.

	Total Product Cost (INR)	Total Sales (INR)	Total Profit (INR)
1	81,11,200.00	1,15,18,500.00	34,07,300.00

2. **Monthly Sales Volume Analysis** – Identified the month with the highest sales volume to understand peak performance periods and seasonal trends.

month	Total Sales (INR)
Feb	23,73,500.00

3. **Top 5 Best-Selling Products Analysis** – Identified the five products with the highest units sold to determine the most in-demand items and overall sales leaders.

	Product	Total Sales (INR)
1	Mouse	70,000.00
2	Chair	2,22,000.00
3	Headphones	2,90,500.00
4	Table	9,15,000.00
5	Tablet	12,75,000.00

4.Category-Wise Profit Contribution Analysis – Determined which category generates the highest total profit to identify the most profitable business segment

	category	Total Profit (INR)
1	Electronics	21,39,000.00
2	Furniture	11,08,800.00
3	Accessories	1,59,500.00

5.Loss-Making Product Analysis – Identified whether any products are generating negative profit to detect loss-making items and evaluate cost or pricing issues

	Product	Category	Total Units Sold at Loss	Total Loss Amount (INR)

Ans : NO

6. Top 2 States by Region (Sales Analysis) – Identified the top two states in each region based on total sales amount to determine the highest revenue-contributing locations within every region.

	ranking	Region	State	Sales Amount (INR)
1	1	East	Bihar	11,01,000.00
2	2	East	Odisha	9,79,500.00
3	1	North	Delhi	11,43,000.00
4	2	North	Punjab	6,02,000.00
5	1	South	Karnataka	9,52,500.00
6	2	South	Tamil Nadu	8,51,500.00
7	1	West	Rajasthan	12,40,000.00
8	2	West	Gujarat	11,21,500.00

7. Regional Profitability Analysis – Determined which region has the highest average profit percentage to identify the most profitable geographic area.

Region	Avg Profit(%)
West	35.57

8. Category-Wise Pricing Analysis – Compared the average unit price and average unit cost for each category to evaluate pricing strategy and profit margins across product segments.

	Category	Avg Selling Price	Avg Production Cost	Avg Markup (Profit per Unit)
1	Electronics	27,122.22	19,700.00	7,422.22
2	Accessories	2,250.00	1,250.00	1,000.00
3	Furniture	14,000.00	9,552.94	4,447.06

9. Top Customer Spending Analysis – Identified the top 10 customers based on their total spending to recognize high-value customers and understand revenue contribution concentration.

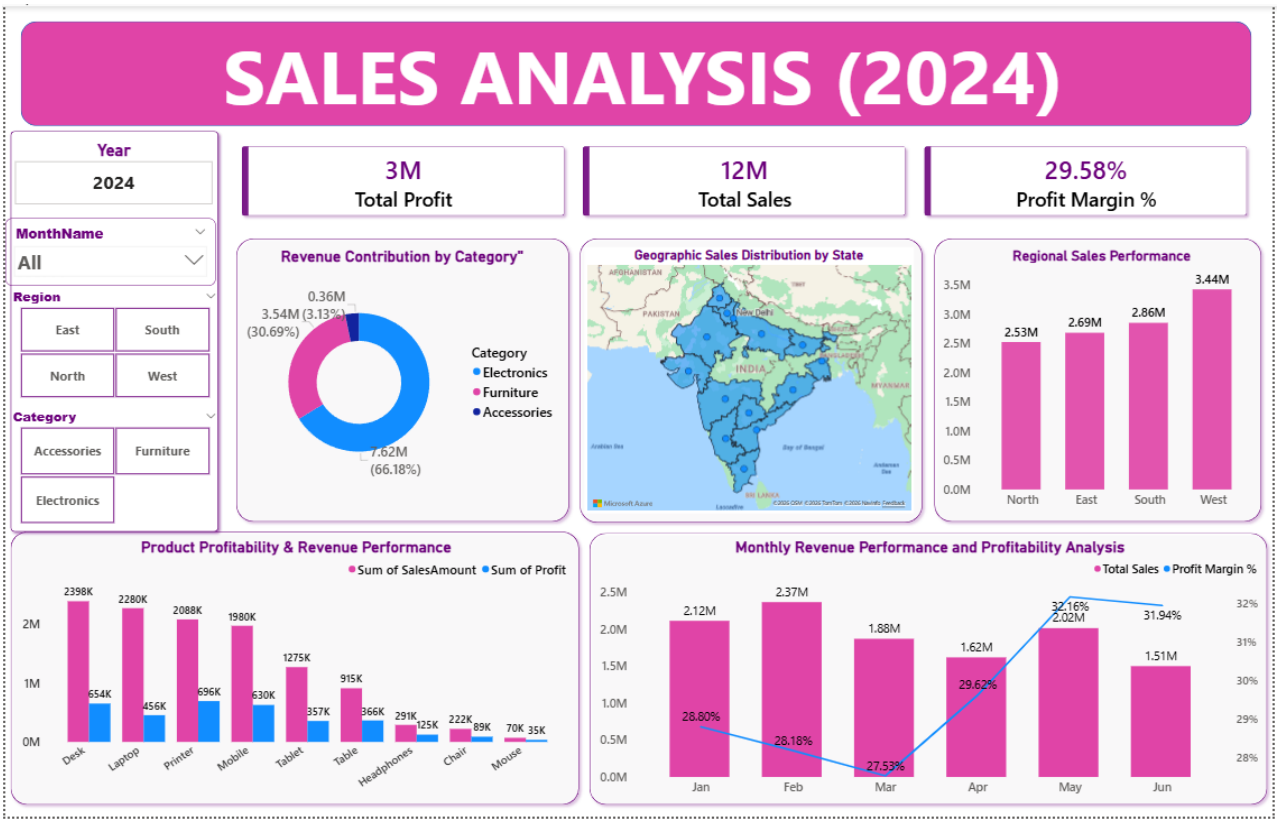
	CustomerName	Total Orders	Total Spending (INR)
1	Rahul	28	20,12,000.00
2	Suresh	19	14,17,000.00
3	Meena	21	12,94,500.00
4	Kiran	24	12,04,500.00
5	Pooja	22	11,20,000.00
6	Anita	11	10,66,500.00
7	Arjun	17	9,54,500.00
8	Neha	21	9,10,000.00
9	Divya	20	8,63,500.00
10	Ravi	17	6,76,000.00

10. Sales vs. Profit Analysis – Compared the top 10 orders by sales amount and profit to evaluate whether high sales always translate into high profit, highlighting discrepancies between revenue and profitability.

	OrderID	Product	Category	Sales Amount	Profit Amount	Profit(%)
1	3103	Laptop	Electronics	3,60,000.00	72,000.00	20
2	3055	Laptop	Electronics	2,40,000.00	48,000.00	20
3	3168	Laptop	Electronics	2,40,000.00	48,000.00	20
4	3074	Laptop	Electronics	1,80,000.00	36,000.00	20
5	3137	Laptop	Electronics	1,80,000.00	36,000.00	20
6	3156	Laptop	Electronics	1,80,000.00	36,000.00	20
7	3173	Laptop	Electronics	1,80,000.00	36,000.00	20
8	3044	Tablet	Electronics	1,50,000.00	42,000.00	28
9	3136	Tablet	Electronics	1,50,000.00	42,000.00	28
10	3010	Mobile	Electronics	1,32,000.00	42,000.00	31.82

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

● **Focus on High-Profit Categories** – Prioritize Electronics in marketing and promotions while boosting Furniture and Accessories through bundles or special offers.

● **Regional Sales Optimization** – Launch targeted campaigns in North and East regions to increase revenue and balance regional sales performance.

● **Product-Level Strategy** – Emphasize top-performing products like Desk, Laptop, and Printer; review or discount low-performing items like Chair, Mouse, and Headphones.

● **Seasonal Promotions** – Leverage peak months (February and May) for promotional campaigns, inventory planning, and special offers to maximize revenue.

● **Profit Margin Management** – Improve margins on mid-performing products by optimizing pricing or reducing costs to increase overall profitability.

● **Cross-Selling & Upselling** – Encourage customers to purchase Accessories with Electronics or Furniture to increase basket size and revenue.